

Shangyu Xing

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[[Profile Page](#), [Google Scholar](#), [GitHub](#)]



EDUCATION

Nanjing University

Master of Computer Science

Natural Language Processing Group (advised by Prof. Xinyu Dai)

Average Score: 88 / 100

Nanjing, China

Sep 2023 - Jun 2026 (expected)

Nanjing University

Bachelor of Computer Science

Average Score: 91 / 100 (Top 5%)

Nanjing, China

Sep 2019 - Jun 2023

RESEARCH INTERESTS

Multimodality: Multimodal Alignment / Multimodal Large Language Models

Post-Training for LLM: RL for (Multimodal) Large Language Models

PUBLICATIONS

Totally 9 papers completed (including 3 in review), first author on 4 of them. 1 additional working paper.

Multimodal LLM:

- [1] **Shangyu Xing**, Changhao Xiang, Yuteng Han, Yifan Yue, Zhen Wu, Xinyu Liu, Zhangtai Wu, Fei Zhao, Xinyu Dai. *GePBench: Evaluating Fundamental Geometric Perception for Multimodal Large Language Models*. Submitted to **ICLR' 2026**, under review.
- [2] Fei Zhao, Chengqiang Lu, Yufan Shen, Qimeng Wang, and Yicheng Qian, Haoxin Zhang, Yan Gao, Yao Hu, Zhen Wu, Dingjie Song, **Shangyu Xing**, Xinyu Dai. *RealBench: A Chinese Multi-image Understanding Benchmark Close to Real-world Scenarios*. **EMNLP' 2025** (findings).
- [3] Fei Zhao, Taotian Pang, Chunhui Li, Zhen Wu, Junjie Guo, **Shangyu Xing**, Xinyu Dai. *AlignGPT: Multi-modal Large Language Models with Adaptive Alignment Capability*. Preprint.

Post-Training for (Multimodal) LLM:

- [4] **Shangyu Xing**, Fei Zhao, Zhen Wu, Tuo An, Weihao Chen, Chunhui Li, Jianbing Zhang, Xinyu Dai. *EFUF: Efficient Fine-grained Unlearning Framework for Mitigating Hallucinations in Multimodal Large Language Models*. **EMNLP' 2024 (main)**.
- [5] **Shangyu Xing**, Siyuan Wang, Chenyuan Yang, Xinyu Dai, Xiang Ren. *Lookahead Tree-Based Rollouts for Enhanced Trajectory-Level Exploration in Reinforcement Learning with Verifiable Rewards*. Submitted to **ICLR' 2026**, under review.
- [6] Yiyang Zhou, Zhaoyang Wang, Tianle Wang, **Shangyu Xing**, Peng Xia, Bo Li, Kaiyuan Zheng, Zijian Zhang, Zhaorun Chen, Wenhao Zheng, Xuchao Zhang, Chetan Bansal, Weitong Zhang, Ying Wei, Mohit Bansal, Huaxiu Yao. *AnyPrefer: An Agentic Framework for Preference Data Synthesis*. **ICLR' 2025**.

Traditional NLP/Multimodal Tasks:

- [7] **Shangyu Xing***, Fei Zhao*, Zhen Wu, Chunhui Li, Jianbing Zhang, and Xinyu Dai. *DRIN: Dynamic Relation Interactive Network for Multimodal Entity Linking*. **ACMMM' 2023**.

- [8] Wen-Shu Fan, Su Lu, **Shangyu Xing**, Xin-Chun Li, De-Chuan Zhan. *Maximizing the Effectiveness of Larger BERT Models for Compression*. **ACL’ 2025 (main)**.
- [9] Fei Zhao, Chunhui Li, Zhen Wu, **Shangyu Xing**, Xinyu Dai. *Learning from Different text-image Pairs: A Relation-enhanced Graph Convolutional Network for Multimodal NER*. **ACMMM’ 2022**.

RESEARCH EXPERIENCES

Multimodal LLMs

- **Pretraining – AlignGPT [3]:** Create distinct alignment vectors for differently aligned text-image pairs during pretraining, and allocate them to various subtasks in finetuning and inference.
- **Benchmarking – GePBench [4]:** Construct a large-scale dataset for evaluating fundamental geometric perception capabilities in MLLMs. We find that even the advanced models struggle, and models trained on our dataset exhibit stronger performance on real-world tasks.
- **Cross-Domain Applications – Paleontological Fossils Captioning [working paper]:** Generates precise, standardized textual descriptions for species-specific fossils, supporting classification and education. Led a 6-member team to build a captioning system integrating MLLM and vision expert tools. System exceeds GPT-4o by 20% in domain evaluation; targeting submission to Nature/Science sub-journals.

Post-Training for (Multimodal) LLMs

- **Hallucination Mitigation – EFUF [1]:** Leverage external expert knowledge to reinforce the alignment between language and vision, thereby reducing multimodal hallucinations with no manually annotated data and minimal computational resources.
- **Lookahead Tree-based group sampling for GRPO [5]:** Constructs tree structures per sequence group, branching on high-probability tokens and dynamically pruning similar branches during generation, significantly enhancing diversity and exploration. Achieves 131% policy learning acceleration and +4.2% performance with -10% inference-time cost across 5 reasoning datasets.

INTERNSHIP

UNC Chapel Hill

North Carolina, USA

Research Intern led by Huaxiu Yao (Online)

Jun 2024 - March 2025

- Designed preference data synthesis algorithm with feedback on vision-language model [2]

University of Southern California

California, USA

Research Intern led by Sean (Xiang) Ren (Online)

April 2025 – Present

- Studying Lookahead Tree-based group sampling for GRPO [5]

SKILLS

Programming: Python, C/C++, Assembly, Java

Frameworks: PyTorch, Hugging Face, Deepspeed, Megatron-LM

Tools: Linux, Docker, Git, LaTeX

Languages: Chinese (Native), English (TOEFL 105)

AWARDS

National Scholarship (2020)

Tencent Scholarship (2021, 2024)

Outstanding Student Model of Nanjing University (TOP 1%) (2022)

First-class Academic Scholarship for Master’s Students of Nanjing University (2024)